

Tutorial Letter 103/0/2026

Programming: Contemporary Concepts COS2614

Year Module(s)
Computer Science

IMPORTANT INFORMATION

Please register on myUnisa, activate your myLife e-mail account and make sure that you have regular access to the myUnisa module website, COS2614-2026-Y, as well as your group website.

Note: This is a fully online module. It is, therefore, only available on myUnisa.

BARCODE

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1 INTRODUCTION

Dear Student

This assignment reinforces your understanding of Object-Oriented Programming (OOP) . You will apply encapsulation, inheritance, polymorphism, abstraction, file handling, and menu-driven program design.

2 ASSIGNMENT

2.1 Outcomes covered in the assignment

The following outcomes will be covered in Assignment number: 2

- Demonstrate an understanding of inheritance in an object-oriented programming (OOP) language
- Demonstrate an understanding of polymorphism in an object-oriented programming (OOP) language

2.2 Assignment information

Details of assignment can be found in the table below:

Assignment weight contribution towards final mark:	30%
Method of assessment (project/takehome/written/portfolio)	Project
Total marks:	100
The assignment is: (compulsory/mandatory/elective)	Compulsory
The assignment will/not be proctored using IRIS	Not
Turn-it-in will be used for projects/takehome/written assignments	Not

The date for submission can be found on myUNISA.

2.3 Assignment submission instructions

- Unisa, as a comprehensive open distance e-learning institution (CODeL), is moving towards becoming an online institution. You will therefore see that all your study material, assessments and engagements with your lecturer and fellow students will take place online. We use myUnisa as our virtual campus.

- The myUnisa virtual campus will offer students access to the myModules site, where learning material will be available online and where assessments should be completed. This is an online system that is used to administer, document, and deliver educational material to students and support engagement between academics and students.
- The myUnisa platform can be accessed via <https://my.unisa.ac.za>. Click on the myModules 2024 button to access the online sites for the modules that you are registered for.
- The university undertakes to communicate clearly and as frequently as is necessary to ensure that you obtain the greatest benefit from the use of the myModules learning management system. Please access the announcements on your myModules site regularly, as this is where your lecturer will post important information to be shared with you.
- When you access your myModules site for the module/s you are registered for, you will see a welcome message posted by your lecturer. Below the welcome message you will see the assessment shells for the assessments that you need to complete. Some assessments may be multiple choice, some tests, others written assessments, some forum discussions, and so on. All assessments must be completed on the assessment shells available on the respective module platforms.
- To complete quiz assessments, please log on to the module site where you need to complete the assessment. Click on the relevant assessment shell (Assessment 1, Assessment 2, etc.). There will be a date on which the assessment will open for you. When the assessment is open, access the quiz online and complete it within the time available to you. Quiz assessment questions are not included in this tutorial letter (Tutorial Letter 101) and are only made available online. You must therefore access the quiz online and complete it online where the quiz has been created.
- It is not advisable to use a cell phone to complete the quiz. Please use a desktop computer, tablet or laptop when completing the quiz. Students who use a cell phone find it difficult to navigate the **Online Assessment** tool on the small screen and often struggle to navigate between questions and successfully complete the quizzes. In addition, cell phones are more vulnerable to dropped internet connections than other devices. **If at all possible, please do not use a cell phone for this assessment type.**
- For written assessments, please note the due date by which the assessment must be submitted. Ensure that you follow the guidelines given by your lecturer to complete the assessment. Click on the submission button on the relevant assessment shell on myModules. You will then be able to upload your written assessment on the myModules site of the modules that you are registered for. Before you finalise the upload, double check that you have selected the correct file for upload. Remember, no marks can be allocated for incorrectly submitted assessments.

2.4 Questions for Assignment 1.

Objective

Develop a console-based Vehicle Rental Management System using Qt 6 (Qt Core). The goal is to demonstrate Object-Oriented Programming (OOP) principles (encapsulation, inheritance, polymorphism, abstraction), menu-driven interaction, file handling, and rental/return state management.

Requirements

1. Class Design (OOP Concepts) – 40 Marks

Base Class: Vehicle (Abstract)

- Private data members:

QString id
 QString brand
 QString model
 double pricePerDay
 bool isRented

- Required Member Functions:

Getters and setters for all attributes (encapsulation)
 Pure virtual function: virtual QString typeName() const = 0;
 Virtual function to display information (e.g., toDisplayString() or displayInfo())

- Derived Classes:

Car : public Vehicle

Additional attributes: int numberOfDoors; int numberOfSeats;

Motorcycle : public Vehicle

Additional attribute: int engineCapacityCC

2. Console Interface (Menu Driven) – 30 Marks

Use QTextStream for console input/output.

- Menu options must include:

Add vehicle (choose type: Car or Motorcycle)
 Search vehicle by ID
 Display all vehicles
 Display available vehicles only
 Rent vehicle

Return vehicle
Exit

- Interface Requirements:

Validate numeric input
Prevent duplicate or empty IDs
Provide clear success/error messages

3. File Handling (Data Persistence) – 20 Marks

Use Qt file classes (QFile and QTextStream).

- The program must:

Load vehicles from file when the application starts
Save all vehicles to file when program ends.

- Suggested file format (pipe-separated):

Car|ID|Brand|Model|Price|IsRented|Doors
Motorcycle|ID|Brand|Model|Price|IsRented|EngineCC

4. Rent/Return Functionality – 10 Marks

- Rent Vehicle:

Enter vehicle ID
Prevent renting if already rented
Update rental status and save to file

- Return Vehicle:

Enter vehicle ID
Prevent return if vehicle is not rented
Update rental status and save to file

Submission Guidelines

- Submit a compressed .zip file containing:

All .cpp and .h files
CMakeLists.txt
README.txt with build and run instructions
Any required data file (if applicable)
Screenshots of test runs of all working functionality.

Marking Rubric

Class Design (OOP) – 40 Marks

Console Interface – 30 Marks

File Handling – 20 Marks

Rent/Return Functionality – 10 Marks

Total – 100 Marks

2.5 Proctoring and plagiarism

Since 2020 Unisa conducts all its assessments online. Given stringent requirements from professional bodies and increased solicitations of Unisa's students by third parties to unlawfully assist them with the completion of assignments and examinations, the University is obliged to assure its assessment integrity through the utilisation of various proctoring tools: Turnitin, Moodle Proctoring, the Invigilator App and IRIS. These tools will authenticate the student's identity and flag suspicious behaviour to assure credibility of students' responses during assessments. The description below is for your benefit as you may encounter any or all of these in your registered modules:

Turnitin is a plagiarism software that facilitates checks for originality in students' submissions against internal and external sources. Turnitin assists in identifying academic fraud and ghost writing. Students are expected to submit **typed** responses for utilisation of the Turnitin software.

The **Moodle Proctoring tool** is a facial recognition software that authenticates students' identity during their Quiz assessments. This tool requires access to a student's **mobile or laptop camera**. Students must ensure their camera is activated in their browser settings prior to their assessments.

The **Invigilator "mobile application-based service** does verification" of the identity of an assessment participant. The Invigilator Mobile Application detects student dishonesty-by-proxy and ensures that the assessment participant is the registered student. This invigilation tool requires students to download the app from their Play Store (Google, Huawei and Apple) on their **mobile devices** (camera enabled) prior to their assessment.

IRIS Invigilation software verifies the identity of a student during assessment and provides for both manual and automated facial verification. It has the ability to record and review a student's assessment session. It flags suspicious behaviour by the students for review by an academic administrator. IRIS software requires installation on students' **laptop devices** that are enabled with a webcam.

Students who are identified and flagged for suspicious dishonest behaviour arising from the invigilation and proctoring reports are referred to the disciplinary office for formal proceeding.

Please note:

Students must refer to their module assessment information on their myModule sites to determine which proctoring or invigilation tool will be utilised for their formative and summative assessments.

3 ACADEMIC DISHONESTY

3.1 Plagiarism

Plagiarism is the act of taking the words, ideas and thoughts of others and presenting them as your own. It is a form of theft. Plagiarism includes the following forms of academic dishonesty:

- Copying and pasting from any source without acknowledging the source.
- Not including references or deliberately inserting incorrect bibliographic information.
- Paraphrasing without acknowledging the original source of the information.

3.2 Cheating

Cheating includes, but is not limited to, the following:

- Completing assessments on behalf of another student, copying the work of another student during an assessment, or allowing another student to copy your work.
- Using social media (e.g. WhatsApp, Telegram) or other platforms to disseminate assessment information.
- Submitting corrupt or irrelevant files, this forms part of examination guidelines
- Buying completed answers from so-called “tutors” or internet sites (contract cheating).

3.3 For more information about plagiarism, follow the link below:

<https://www.unisa.ac.za/sites/myunisa/default/Study-@-Unisa/Student-values-and-rules>

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